

Md Shakil Arman

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Professional Summary

Ph.D. candidate in Industrial Engineering at Texas A&M University, specializing in advanced manufacturing, powder processing, and materials characterization. Experienced in the design of experiments and statistical analysis to reduce process variability and improve dimensional accuracy, surface quality, and manufacturing reliability. Seeking opportunities in both academia and industry—advancing research and mentorship while driving process innovation and quality improvement.

Education Details

Ph.D in Industrial Engineering.

Texas A&M University, College Station, Texas, USA.

Advisor: Dr. Zhijian Pei.

May 2027 (Expected)

M. Sc. in Manufacturing Engineering.

The University of Texas Rio Grande Valley, Texas, USA.

Advisor: Dr. Jianzhi Li.

2021

B. Sc. in Industrial & Production Engineering

Khulna University of Engineering & Technology ([KUET](#)), Bangladesh.

2015

Research Experience

Graduate Research Assistant- Department of Industrial and Systems Engineering, TAMU.

(2021-Present)

Manufacturing Process Development & Powder Processing

- Perform binder jetting 3D printing with diverse feedstock powders, including ceramic powder (alumina, silicon carbide, boron nitride, zeolites), biomass powder (algae), biowaste-derived powder (eggshell powder), and composite powder.
- Investigate the effect of the powder packing rate on the dimensional accuracy and structural integrity of binder-jetted components.
- Develop granulated powders from eggshell and alumina using spray freeze drying, analyzing the effects of air pressure, slurry feed rate, and solid loading on granule shape, size, and flowability.
- Explore the influence of critical parameters (layer thickness, compaction thickness, binder saturation, dispense volume, roller speed, and temperature) on green part dimensional accuracy, structural integrity, and density.
- Investigate powder segregation during powder mixing and spreading.
- Operate and maintain ExOne Innovent+ and ComeTrue T10 printers, including troubleshooting in collaboration with ExOne technical support.
- Perform basic grinding and lapping operations for surface finishing and dimensional control.

Powder Characterization

- Conduct particle size distribution measurements using a laser scattering analyzer.
- Assess powder flowability through apparent density, angle of repose tests, and tap density with a Hall flowmeter and tap density tester.
- Evaluate binder–powder compatibility across multiple powders.

Experimental Design & Statistical Analysis

- Design experiments using factorial design, randomization, and blocking to ensure reliable results.
- Apply statistical methods (ANOVA, t-tests, F-tests, and paired comparisons) to analyze relationships between process settings and part properties.

Material Characterization & Part Evaluation

- Utilize analytical techniques (Archimedes' method, XRD, SEM, EDS, BET, and mechanical testing) to characterize manufactured parts, including density, crystallinity, surface morphology, porosity, surface area, and compressive strength.

Graduate Research Assistant-Department of Manufacturing and Industrial Engineering - UTRGV.
(2020-2021)

Electrohydrodynamic (EHD) Printing

- Analyzed operating conditions of pulse-electric-field-assisted EHD printing of sodium alginate bioink using design of experiments (DOE).
- Demonstrated that increasing frequency and reducing offset voltage improved droplet rate and spacing while influencing drop stability.

Laser-Induced Forward Transfer (LIFT) Printing

- Investigated bioprinting of retina neuron cells using LIFT.
- Identified collagen I as a suitable bioink for photoreceptor cells and optimized printing strategy for layer-by-layer deposition.
- Evaluated cell viability and morphology in bioprinted constructs.

Co-Axial Extrusion Bioprinting

- Designed and developed a co-axial extrusion bioprinting system using multi-needle/syringe configurations and syringe pumps.
- Fabricated hollow microvessels from two different biomaterials to mimic vascular capillaries.

Tissue Culture System

- Reviewed and evaluated customized bioreactor designs for vascular tissue.
- Investigated criteria for culturing functional vascular tissue under physiologically relevant conditions.

Teaching Experience

Graduate Teaching Assistant- Department of Engineering Technology and Industrial Distribution-TAMU.
(September 2021 to Present)

- **MMET-402- Non Destructive Testing-(2022-present)**
Assist students in performing various non-destructive methods, such as penetrant testing, eddy current testing, Ultrasound and phased array testing, X-ray testing, etc., to identify cracks and defects in metal parts. Assist the course instructor in formulating the exam questions on Canvas, grading the exam sheets, etc.
- **MMET-361- Product Design and Solid Modeling-(2021-2022)**

Taught solid modeling using CREO (computer-aided design software). Assisted the course instructor in grading the lab and exam sheet.

Graduate Teaching Assistant-Manufacturing and Industrial Engineering Department- UTRGV.
(2020-2021)

Helped the professor in the Manufacturing Process Lab. Explained the process of cold rolling, powder metallurgy, and injection molding, and helped the students in conducting the lab experiment.

Job Experience

Executive- Total Quality Management- KDS Steel. 255, Nasirabad I/A, Chittagong-4211, Bangladesh
(From 2016 to 2019)

Responsibilities

- **Total Productive Maintenance:** Identified equipment Fuguai (abnormalities) and rectified them through the mechanical, electrical, or operation team. Monitored preventive maintenance work and compared monthly improvement work.
- **Lean Manufacturing:** Identified muda (waste) of different process lines and optimized them. Monitored 5S works in the plant. Prepared a time and motion study according to lot-wise hot rolled coils (Input raw materials) to make a standard time of completion for every lot.
- **OEE (Overall Equipment Efficiency) Analysis:** Calculated the Mean Time Between Failures (MTBF), Mean Time to Repair (MTTR), Production yield, Equipment availability, Quality yield, i.e., Overall Equipment Efficiency (OEE) of each process line to observe and forecast about each process line.

Skills

Technical skills	: Binder jet 3D printing, EHD 3D printing, LIFT 3D printing, Co-axial extrusion printing, 5S, TPM, TQM.
Programming	: Python, R (Basic)
Utility Software	: CREO, SolidWorks, AutoCAD, TORA, Minitab.

Publications

- Arman, M. S., Khan, F., Pasha, M. M., Sanders, J., & Pei, Z. (2025). Granulation of fine eggshell powder to produce feedstock powder for binder jetting additive manufacturing: A feasibility study. *Materials Today Communications*, 46, 112918. <https://doi.org/10.1016/j.mtcomm.2025.112918>
- Arman, M. S., Xu, B., Tsin, A., & Li, J. (2023). Laser-induced forward transfer (LIFT) based bioprinting of the collagen I with retina photoreceptor cells. *Manufacturing Letters*, 35, 477–484. <https://doi.org/10.1016/j.mfglet.2023.07.005>
- Khan, F., Arman, M. S., Sharma, D., Pei, Z., & Ma, C. (2025). Zeolite types reported in the literature on CO₂ adsorption capacity: Status and gaps. *Separation and Purification Technology*, 378, 134596. <https://doi.org/10.1016/j.seppur.2024.134596>
- Pasha, M. M., Arman, M. S., Pei, Z., Khan, F., Sanders, J., & Kachur, S. (2025). Effects of compaction thickness on density, integrity, and microstructure of green parts in binder jetting additive manufacturing of silicon carbide. *Journal of Manufacturing and Materials Processing*, 9(4), 136. <https://doi.org/10.3390/jmmp9040136>

- Pasha, M. M., Pei, Z., Arman, M. S., Gasdaska, C. J., & Kao, Y. T. (2025). Effects of mixing speed and mixing time on powder segregation during powder mixing for binder jetting additive manufacturing: An experimental study. *Journal of Manufacturing and Materials Processing*, 9(4), 117. <https://doi.org/10.3390/jmmp9040117>
- Khan, F., Sanders, J., Arman, M. S., Pasha, M. M., Kachur, S., & Pei, Z. (2025). Fabrication of SiC–aluminum composites via binder jetting 3D printing and infiltration: A feasibility study. *Journal of Composites Science*, 9(3), 111. <https://doi.org/10.3390/jcs9030111>
- Pasha, M. M., Arman, M. S., Khan, F., Pei, Z., & Kachur, S. (2024). Effects of layer thickness and compaction thickness on green part density in binder jetting additive manufacturing of silicon carbide: designed experiments. *Journal of Manufacturing and Materials Processing*, 8(4), 148. <https://doi.org/10.3390/jmmp8040148>
- Khan F, Arman M. S., Sanders J, Pasha M, Rahman A, Pei Z, Dong T. (2024) Binder Jetting 3D Printing Utilizing Waste Algae Powder: A Feasibility Study. *Intelligent and Sustainable Manufacturing*; 1(2):10016. <https://doi.org/10.70322/ism.2024.10016>
- Rahman, T. T., Arman, M. S., Perez, V., Xu, B., & Li, J. (2021). Analysis of the operating conditions of pulse electric field–assisted EHD for sodium alginate printing using design of experiment approach. *The International Journal of Advanced Manufacturing Technology*, 115(7), 2037–2047. <https://doi.org/10.1007/s00170-021-07042-1>

Conference

- The 35th Annual International Solid Freeform Fabrication Symposium-An Additive Manufacturing Conference.(Aug 11-14, 2024)
-Oral Presentation about granulation of fine eggshell powder into granules for Binder jetting Additive Manufacturing.

Google Scholar link

- <https://scholar.google.com/citations?hl=en&user=bBPzoucAAAAJ>

Honors & Awards

- Presidential Graduate Research Assistantship (PGRA) award (Full tuition fees waived and salary for living expenses) at UTRGV.
- University Scholarship at the undergraduate level for good results every year.

Volunteer Experience

- Mentor approximately ten undergraduate students at Texas A&M University in research and laboratory activities, guiding experimental methods, data analysis, and academic development.
- Served as a student Mentor at I-DREAM4D Consortium (<https://www.idream4d.org/>) at UTRGV (June 3rd-Aug 13th, 2021).
- Engage in student organizations, including Bangladesh Student Association TAMU, UTRGV Muslim Association, and South Asian Association.